Latent Causal Infeasibility for Long-Horizon TAMP

Methodology specification v1.4 · implementation-ready research contract

Intervention-supervised joint relocation for a prescribed target motion

15 October 2026

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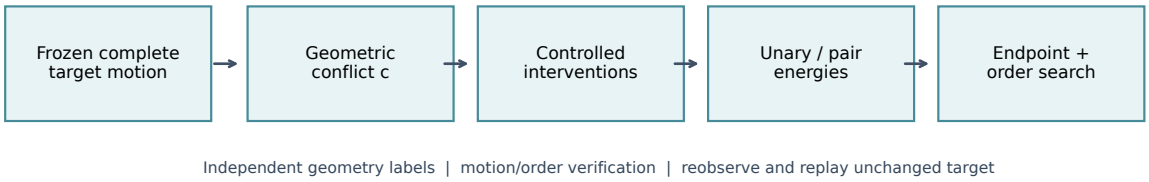
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Status. Normative *design and experimental protocol*, not a record of completed robot experiments. This edition reconciles methodology v1.3 and implementation guide v1.5 with the review of 15 October 2026. The paired *Implementation and Experimental Guide v1.6* contains phase gates, starter configs and code ownership. Both are authoritative only at their declared scopes. Earlier master documents, the

29-case taxonomy, and disc-only examples are historical background and are not permission to change this architecture.

Scientific question. Given observations, supplied object geometry, the current robot state, finite legal resting poses and a **previously validated, now frozen, complete target action**, predict which scene entities obstruct that specific motion and propose a small **executable, move-once** set of relocations that restores it. The model learns compatibility of proposed *endpoints*. Actual robot motion, ordering, contact and replays are checked independently. It does **not** discover latent causal variables, perform unrestricted TAMP, or learn a low-level policy.



End-to-end research interface. This is a conceptual schematic, not measured robot evidence.

1. Immutable research decisions

Dimension	Normative decision
Problem unit	A reference root, one active target, one prescribed action a and complete validated trajectory γ_0, with controlled descendants
2D embodiment	Full fixed-base 3R planar arm, three finite-width links, joint housings, two side-grasp fingers, rigidly attached disc during lateral carry
3D embodiment	Pinned full xArm7 and gripper in a low cupboard, upright cylinders and fixed support-contact planar carry; robot links move freely in 3D
Scene	Initially identical upright cylinders; 8×8 legal resting lattice, pitch 0.08 m, 2D challenge 30–40 obstacles; 3D pilot 6–10
Target motion	Independently verified in populated reference; complete robot/gripper/attachment/contact trace and swept geometry frozen across matched interventions
Observations	GT instance masks, RGB-D, frozen mask-pooled features, supplied known primitive geometry and explicit poses; no object discovery claim
Learner	Trainable candidate-conditioned unary action/workspace and symmetric pair compatibility factors; conditional categorical Boltzmann energy

Dimension	Normative decision
Planning	Categorical assignments to finite physical spots; budget = number of distinct moved objects; no continuous endpoint optimization
Verification	Separate local geometry, terminal arrangement, robot motion/order and simulation replay evidence; three-valued statuses
Repair scope	Each selected object relocated at most once, gripper approach/grasp/carry/release/retreat, persistent moved history and connector to $\gamma_0(0)$
Training	Oracle-labeled local binary factors plus finite-pool joint likelihood; same endpoint semantics for positive and negative examples
Long horizon	Next-action repair and reobservation; future task actions handled by a separate explicitly scripted integration test

Authority rule: this document defines variables, validity, evidence and research claims. Guide v1.6 defines run ordering, nominal configuration defaults, deliverables and stage gates. Resolve a contradiction in both, version it and regenerate impacted labels before implementing further.

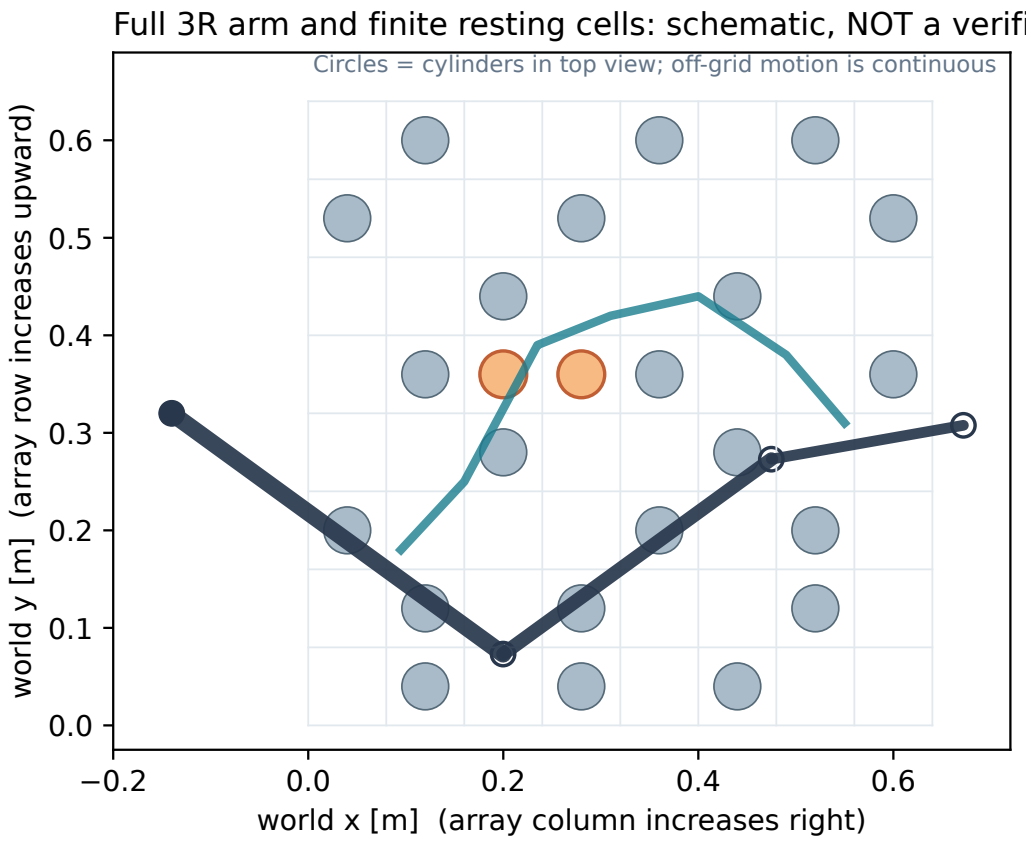
2. Scene, coordinate frames and robot state

Let $\mathcal{O} = \{1, \dots, N\}$ be movable obstacle objects, excluding the active target o_t. Object i has stable ID, known primitive geometry G_i, measured/supplied pose p_i, a support and allowed candidate list. W denotes fixed supports, walls and cupboard bodies; q joint positions, g gripper aperture, $m \in \{\text{free}, \text{attached}\}$ the attachment state. A restorable simulator state is

$$S = (\{p_i, G_i\}_{i=1}^{N}, p_t, q, g, m, W, \eta),$$

where η includes any contact/control/simulation arrays required for exact reset. The learner sees the declared observation/interface, **not** privileged oracle distances, culprit labels or interventions' outcomes.

Use meters, radians, float64 geometry, float32 networks; 3D quaternions serialize as **wxyz**. World 2D x is rightward and y is *backward* on the supporting surface; 3D z points upward, cupboard front is $y = 0$. Array *row* increases with world y (not image-down by default), and array *column* increases with x. Specify row-to-pixel reversal in the renderer. For an 8×8 lattice with pitch $h = 0.08$ m, spot (r, c) has $x = (c + \frac{1}{2})h$, $y = (r + \frac{1}{2})h$, $r, c \in \{0, \dots, 7\}$. The usable 0.64 m square is measured between lattice borders; fixed walls may further restrict legal supports.



A 3R arm, grid cells, obstacle footprints and action curve. Array rows increase upward in world coordinates. Illustrative geometry; not a verified trajectory.

The grid constrains **resting endpoints only**. Intermediate object centers and robot joints trace continuous paths. Only one target is active per action query; others remain collision bodies. Shares of reference roots, sibling trajectories, one-edit twins and nuisance renders must share a split.

2.1 Explicit articulated 2D contract

With fixed base b and three link lengths $L_j > 0$, let $\theta_j = \sum_{k \le j} q_k$. The tool pose is

$$p_{\mathrm{TCP}}(q) = b + \sum_{j=1}^{3} L_j(\cos\theta_j, \sin\theta_j), \qquad \phi_{\mathrm{TCP}} = q_1 + q_2 + q_3.$$

IK for a desired (x, y, ϕ) subtracts the third-link offset, tests **both** two-link elbow branches, reconstructs q_3, and rejects out-of-bounds, colliding or discontinuous candidates. An IK endpoint alone is not a path. The selected object is connected to the tool only after valid side approach and finger closure. During attached carry, maintain the tool–object relative transform, and check every joint housing, link capsule, two fingers, attachment and nonadjacent self-collision. The target skill comprises home-to-pregrasp, side approach, closure/attachment, continuous carry, release/detachment and retreat. A valid connector from the final corrective state to the **exact** stored target start $(q_{\mathrm{start}}, g_{\mathrm{start}}, m = \text{free})$ is required before target replay.

Only `PLANAR_ARM_VERIFIED` establishes articulated 2D kinematic success. `DISC_SURROGATE` and isolated-

tool illustrations cannot label 2D main results.

2.2 3D support-contact transport

The no-lift cupboard skill is **grasped, support-contact planar transport**. A cylinder of height $h_o = 0.075$ m on support top $z_s = 0.005$ m has $z_{\text{center}} = 0.0425$ m and bottom z_s. Its base remains in intended contact with the support throughout supported grasp and carry. It is **not** a freely suspended cylinder and **not** an ungrasped slide. This is a kinematic benchmark convention; it does not establish physical friction or grasp stability.

For an attached body under joint trajectory $q(t)$ require

$$|z_{\text{obj}}(q(t)) - z_*| \le \epsilon_z, \qquad \angle(R_{\text{obj}}(q(t))e_z, e_z) \le \epsilon_{\text{tilt}},$$

for **all** verified intervals, not just endpoints. Upright yaw may vary in the selected side-grasp family; the explicit tool-to-object rigid transform determines allowable yaw. The arm links do **not** have fixed z. The path generator must use constrained IK / projected joint-space samples / constraint-aware motion planning and reject segments that violate height, uprightness or collision bounds.

Pair	3D collision/contact policy
Selected cylinder–support	Intended contact during supported approach, grasp, carry and release; tolerance and support ID logged
Selected cylinder–fingers	Intended contact only inside validated closure/attachment schedule
Selected cylinder–other robot links	Forbidden unless a separately documented grasp fixture explicitly requires otherwise
Selected cylinder–other movable cylinders	Forbidden
Robot–cupboard walls, roof and shelf	Forbidden
Nonselected cylinders–support	Ordinary static resting contact
Nonadjacent robot self pairs	Tested; only documented adjacent joint geometry exclusions allowed

A simulator-only weld after validated closure may be used for **kinematic replay** with an explicitly recorded `attachment_assistance` flag; never use it to excuse impossible approach, closure or transport. Controller, contact-force and hardware success are different fields. No rising over obstacles or vertical escape is allowed.

3. Frozen action and collision envelope

First build a feasible *populated* reference scene S_{ref} and plan a complete continuous action $\gamma_0(t) = (q_0(t), g_0(t), m_0(t))$ with its exact phase/contact schedule. Each accepted 2D trajectory checks all 3R bodies and attached target; each 3D trajectory all xArm7 bodies, fingers, attachment constraints and support contacts. Store robot pose sequence, validated intervals, allowed pairs, simulator/robot hashes and frozen action hash. Paths sampled but never fully validated are `UNRESOLVED`, not clean references.

For passive object checking define $B_{\gamma_0}(t)$ as the union of the target, fingers and relevant robot bodies at time t with phase-specific contact exceptions. Let

$$\mathcal{E}_0 = \bigcup_{t\in[0,1]} B_{\gamma_0}(t), \qquad \mathcal{E}_\delta = \mathcal{E}_0 \oplus B_\delta.$$

Exactly one margin convention: the source-of-truth geometric oracle operates on **uninflated** real primitive shapes/$\mathcal{E}_0$ and demands signed separation $d \geq \delta$ for forbidden body pairs; its numeric boundary tolerance is ϵ_{num}. The learner may receive a sampled representation of $\mathcal{E}_0$ *plus* scalar δ, or $\mathcal{E}_\delta$ plus a flag `already_inflated=true`, never inflate that second representation again. Store both semantics, representation resolution and original geometry provenance.

For an obstacle at candidate pose p_i^k, action compatibility is

$$y_i^a(k) = \mathbf{1}\left[\min_{t\in[0,1]} d(G_i(p_i^k), B_{\gamma_0}(t)) \geq \delta\right],$$

under correct contact exceptions. Clear with certified $d \geq \delta + \epsilon_{\text{num}}$; violating with certified $d < \delta - \epsilon_{\text{num}}$; the gray band is unresolved until refined. An ordinary fixed-action collision can be verified even when the original task would succeed via rerouting. The optional reroute-first baseline solves the **different** no-relocation free-path question and cannot alter a frozen action's label.

Reference-to-failure interventions move existing obstacle IDs (never insert new objects) to distinct available legal resting cells that overlap the frozen sweep **while remaining present-state nonpenetrating**. Save every adjacent one-edit twin $S^{(j-1)} \to S^{(j)}$, the entire five-edit ancestry where relevant and the unmodified γ_0 hash. Diagnostic pose edits are privileged and do not count as executed corrective moves.

Continuous verification: sampled collision-free configurations are not a certificate. Apply analytic sweep checks in 2D or conservative adaptive interval subdivision in 2D/3D with established distance-change bounds. A sufficient endpoint bound is $\min(d(t_j), d(t_{j+1})) - L_d\Delta t/2 \geq \delta$ **only if** L_d is verified. If no conservative bound or continuous collision routine resolves a segment, return `UNRESOLVED` and record its interval and call budget. Pin MuJoCo release and test `mj_geomDistance` against known contact, near miss and penetration cases; a bounded `distmax` answer is not an exact unlimited distance.

4. Endpoint domain and compatibility

For object i define categorical $x_i \in \{0, \dots, K_i\}$, with $0 =$ **its actual current stay pose** and each $k > 0$ referencing a *physical* spot ID and pose p_i^k. Candidate indexes are per-object, **not** global spot IDs. Deduplicate identical pose values. Each candidate must pass support, fixed workspace, target/object occupancy semantics and the declared finite domain. Ghosts substitute a pose for the same ID; they never clone the object. Pair factors test simultaneous hypotheses, including two objects requesting the same cell.

Local semantic labels:

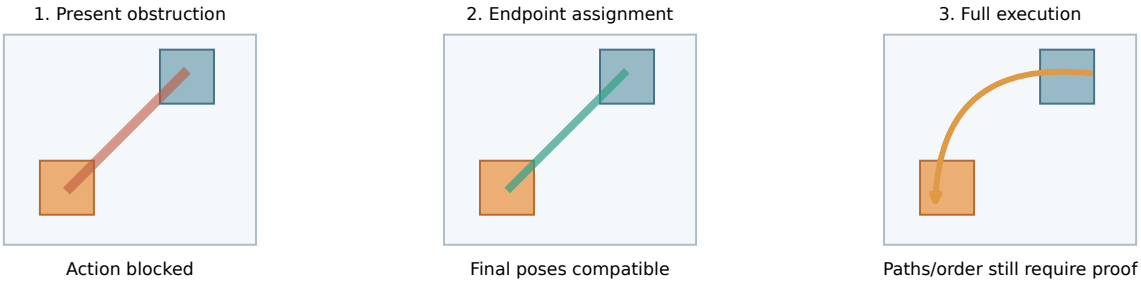
$$y_i^a(k): \text{ candidate clears frozen action}, \qquad y_i^w(k): \text{ candidate has valid support/fixed geometry},$$

$$y^p_{ij}(k,\ell): \text{ objects } i,j \text{ coexist at selected endpoints.}$$

For a definite endpoint assignment, terminal validity factorizes:

$$Y_{\text{term}}(x) = \prod_i y^a_i(x_i) y^w_i(x_i) \prod_{i<j} y^p_{ij}(x_i, x_j).$$

This conjunction is exact only under the independently supported rigid-object, fixed-target-trajectory, pair-collision model. It does not prove a valid *sequence* of robot moves. When factors are tri-valued: **any definite zero means invalid; every factor definite one means valid; otherwise unresolved**. Unresolved is not a negative and must be masked from the training loss. A zero from one factor wins even when unrelated factors are unknown.



Terminal compatibility and full robot execution answer different questions. Schematic only.

5. Local factors, energies and Boltzmann proposals

Frozen visual backbone F_ϕ produces image patch features pooled by the **visible** area of the GT instance mask. For mask fraction w_{ivp},

$$\bar{f}_i = \frac{\sum_{v,p} w_{ivp} F_\phi(I_v)_p}{\sum_{v,p} w_{ivp}}, \quad z_i = f_\theta(\text{stopgrad}\, \bar{f}_i).$$

Zero visible area is **missing data**, never an observed all-zero feature. Log pose provenance; supplied simulator pose is not "recovered from RGB." A candidate token combines z_i, known dimensions, normalized candidate coordinates, support context and relative action/workspace geometry. Recompute candidate-relative geometry for each ghost. Do not feed integer IDs as ordered geometric variables or distance-to-collision *labels* as inputs. For identical cylinders, a geometry-only ablation is mandatory.

The shared unary network emits action and workspace logits $\ell^a_i(k), \ell^w_i(k)$; the shared pair network emits $\ell^p_{ij}(k,\ell)$ with explicit object-exchange symmetry, e.g. average $g(h_i,h_j)$ and $g(h_j,h_i)$. Turn **learned** logits into nonnegative energies:

$$q_f = \sigma(\ell_f), \quad e_f = -\log q_f = \text{softplus}(-\ell_f),$$
$$U_i(k) = e^a_i(k) + e^w_i(k), \quad V_{ij}(k,\ell) = e^p_{ij}(k,\ell), \quad E_\theta(x \mid c) = \sum_i U_i(x_i) + \sum_{i<j} V_{ij}(x_i, x_j).$$

Each factor persists even for stay choices, so no spurious energy drop is obtained by deleting a candidate/object. Pair tables are stored only for $i < j$. This graph is an **undirected compatibility factorization**; it is not a causal DAG, a decoder, a second learned network or a sequence graph. The energy is dimensionless and **not** penetrative depth, a physical success probability or a guaranteed unique causal representation.

For at-most-b moved objects, $M(x) = \{i : x_i \neq 0\}$,

$$P_{\theta,b}(x \mid c) = \frac{\mathbf{1}[|M(x)| \leq b] e^{-E_\theta(x|c)/T}}{Z_{\theta,b}(c)}, \quad T > 0.$$

Inference updates assignments, **never network weights**. For k permitted by the remaining moved budget, the categorical Gibbs conditional uses $H_i(k) = U_i(k) + \sum_{j<i} V_{ji}(x_j, k) + \sum_{j>i} V_{ij}(k, x_j)$ and samples $x_i = k$ with probability proportional to $e^{-H_i(k)/T}$. Use random update order, seeded parallel chains, mixed initializations and deduplication. A sampler's failure to propose a solution is `UNRESOLVED`, not a proof of infeasibility; its samples are proposals to be checked with the same independent oracle used for baselines.

6. Causal evidence: detect, intervene, attribute

Define the direct blocker set for the **frozen target sweep** as

$$D(S; \gamma_0) = \{i : y_i^a(0) = 0\}.$$

The simulator allows controlled $do(p_i \leftarrow p_i^k)$ and joint endpoint edits while restoring all other physical state and keeping γ_0 constant. The action-success effect is $\Delta_I = F_{\text{fixed}}(S^I; \gamma_0) - F_{\text{fixed}}(S; \gamma_0)$. Keep per-entity conflict contribution $c(S, \gamma_0) = [c_1, \ldots, c_N]$ and intervention deltas $d_I = c(S^I, \gamma_0) - c(S, \gamma_0)$ as distinct from the binary total outcome. Several independently blocking objects may each have **zero individual effect on overall action success** even though each object's interference disappears. Plot both total outcome and per-object changes.

Diagnostic removal ($do(\text{present}_i = 0)$), **diagnostic presence**, and **physical relocation** answer different counterfactuals. Removing an immovable wall can diagnose its geometric involvement but is not a legal repair; cause entity and moved repair target need not coincide in general. For the current benchmark, only movable obstacles undergo the prescribed relocation. A minimum **removal repair set** is not a minimum **failure-preserving presence set**, and inclusion-minimal is not minimum-cardinality. Deleting nonnegative learned energy factors is **not** causal evidence. Any causal claim is operational and conditional on the simulator and frozen action, not latent-variable identifiability.

7. New reference-grid theorem and the genuine optimization problem

The terminal restoration lemma (normative). Assume: (i) the frozen target action was verified with *all* N obstacles at mutually distinct supported reference cells; (ii) failure is induced exclusively by moving an existing direct-blocker set D from reference cells into obstructing cells, without changing other objects or the target; (iii) every obstacle is an identical footprint; (iv) **all supported cells** remain candidate endpoints; (v) each pair of distinct cells automatically clears the required obstacle-obstacle

separation; and (vi) the terminal verifier checks the frozen action and ordinary pairwise collisions. Then the original reference cells of the displaced blockers form a valid terminal assignment. Each direct blocker must move at least once, because its obstructing stay pose intersects the fixed envelope. Hence

$$\boxed{b^*_{\text{terminal}} = |D|.}$$

For the initial discs, pitch 0.08 m and $2r + \delta = 2(0.024) + 0.005 = 0.053$ m ensure **distinct** grid centers are mutually compatible. A current failure with $|D| = 5$ therefore has an unrestricted terminal five-move solution: restoring the five reference poses. An occupied preferred destination can invalidate **one assignment** but cannot force a sixth *terminal* relocation under these assumptions. Nor can the ordinary reference-to-failure construction produce an unrestricted no-terminal-solution case. These conclusions apply to endpoints, **not** to full articulated movement or robot-state connectors.

A six-relocation **executable** outcome with five direct blockers must be justified by measured motion/access/order restrictions (including a genuinely necessary staging occupant) **or** an explicitly restricted domain that excludes the reference restoration route. If neither is established, the scene is not a validated six-relocation example. No "six is necessary" label may be inferred from one failed five-move randomized search.

7.1 Explicitly restricted-domain controls

Family `F8` / fixture `T07` must be **separately constructed**: start from a reference/action pair, then define and publish a restricted legal candidate set excluding the necessary safe return cells and any alternative full matching. Establish emptiness by exact Hall/matching certificate (when pairing is only cell uniqueness) or complete finite endpoint enumeration / sound solver certificate. Mark `domain_policy=RESTRICTED`, log excluded cells **and rationale**, and include its **ALL_SUPPORTED** counterpart in the same root split. It is logically invalid to treat the unrestricted counterfactual as terminal-infeasible. A motion-inaccessible cell remains endpoint-valid; do not silently remove it from the terminal domain to manufacture an endpoint failure.

7.2 Minimum cost *terminal* matching baseline (A0)

On the initial grid, distinct supported spots are pair-compatible. Construct bipartite edges from *each* object (left) to **every physically supported, action-clear, fixed-valid cell** (right). Edge cost is 0 if the object stays at that physical cell and 1 otherwise. Require one distinct right cell per object. Min-cost maximum-cardinality matching minimizes the **number of relocated objects at terminal time**. The same procedure must use **exactly the permitted domain** in restricted-domain controls. It is not valid if different cells can geometrically overlap for heterogeneous/wider footprints, non-grid poses or coupled support constraints; in those variants use explicit pairwise constraints and a general discrete solver.

One matching may be motion-impossible. A0 therefore attempts **k-best distinct assignments** in nondecreasing moved-object count/cost, with deterministic no-good exclusions, and calls the **same** move-once order/motion verifier as every learned method until an agreed witness/time/call budget is exhausted. For complete small domains, all terminal assignments and allowed robot orders may be enumerated; for large domains, finite search budget exhaustion is `UNRESOLVED` without completeness. Do **not** brute-force $N = 35$, $K = 63$ with $b \le 5$; $\sum_{m=0}^{5} \binom{35}{m} 63^m \approx 3.23 \times 10^{14}$ choices before pruning.

Keep **ALL_SUPPORTED** A0 as a mandatory benchmark control even if an attractive restricted candidate list makes pair interactions look harder. Train/evaluate model on **the domain actually declared for that experiment**, show both domain policies side by side, and never infer "pair factors necessary" simply from unique-cell assignment under wide grid spacing.

8. Executable repair, ordering, history and proof scope

A sequence π contains at most one relocation per selected obstacle, performed with actual planned robot approach, grasp, fixed-height carry, release and retreat. After each move, update actual scene and robot/gripper state. At episode start $H = \emptyset$; after a move object identity i is inserted into H and its candidate domain becomes stay at the **achieved** pose. Reobservation recalculates inputs and energies but preserves H, root/action hash, spent budget and all physical objects. Do not reset history when recursing or switching query IDs within one repair episode.

For a proposed assignment, first check independent terminal compatibility, then search orders and collision-checked paths; then verify connector to the pinned γ_0 start and **replay the unchanged** γ_0. A plausible endpoint assignment is not a replay witness. The target cannot be silently replanned. Scope and statuses:

Stage	Evidence / permissible claim
`LOCAL_GEOMETRY`	Factor validity for a single candidate, with explicit contact and clearance conventions
`TERMINAL_ONLY`	Full assignment coexists and clears frozen action at final poses
`PLANAR_ARM_VERIFIED`	Full 3R links + fingers + attachment + ordered moves + connector + replay validated kinematically
`FULL_ROBOT_VERIFIED`	Full 3D xArm7 + gripper + attached object, cupboard contacts, ordered moves, connector and kinematic replay
`SIM_REPLAY_SUCCESS`	Separately successful dynamic/controller simulation under declared assistance and contact mode
`HARDWARE_SUCCESS`	Separately measured real robot execution; never implied by kinematic status

Every query status is exactly `VERIFIED_FEASIBLE`, `CERTIFIED_INFEASIBLE` (with explicit **finite/sound proof scope**) or `UNRESOLVED`. Use a separate `OUT_OF_SCOPE` eligibility flag. A stochastic planner timeout, numerical error, unverified interval, IK failure under incomplete branch/search, unavailable robot asset or controller failure **cannot** be reported as certified global physical infeasibility.

A sound direct-blocker lower bound $b^*_{\text{exec}} \geq |D|$ and a replay-verified witness with **exactly** $|D|$ relocations together prove the scoped move-once optimum; no separate exhaustive searches at lower budgets are needed. Otherwise report b_{LB}, best verified b_{UB} (if any), and `MINIMUM_CERTIFIED` only if proof closes. "Smallest found" is not minimum. For general restricted domains, an exact terminal matching lower bound is also valid for executable sequences, and can be combined with any other **sound** lower bound. An upper bound is legitimate only with complete executable witness.

9. Labels and losses

Local factor targets $y_f \in \{0, 1\}$ come from the independent oracle, not teacher preference or learned energy. Use BCE-with-logits, balanced factor-type sampling and naturally distributed test evaluation:

$$\mathcal{L}_{\text{factor}} = \frac{1}{|B|} \sum_{f \in B} \text{BCEWithLogits}(\ell_f, y_f).$$

Deduplicate a finite candidate pool C for one scene/action; independently label terminal valid $C^+ \subseteq C$ while masking unresolved assignments. When positives and negatives both exist,

$$\widetilde{P}_\theta(x \mid C, c) = \frac{e^{-E_\theta(x|c)/T_{\text{train}}}}{\sum_{u \in C} e^{-E_\theta(u|c)/T_{\text{train}}}}, \qquad q_C(x) = \frac{\mathbf{1}[x \in C^+]}{|C^+|},$$
$$\mathcal{L}_J = -\sum_{x \in C} q_C(x) \log \widetilde{P}_\theta(x \mid C, c), \qquad \mathcal{L} = \mathcal{L}_{\text{factor}} + \lambda_J \mathcal{L}_J.$$

For independent energy variables, $\partial \mathcal{L}_J / \partial E(x) = (q_C(x) - \widetilde{P}_\theta(x))/T_{\text{train}}$. **All terminal-valid pool examples are positives**, whether or not physically executed; an unresolved motion search does not flip the terminal label. Skip joint term for all-invalid pools; all-valid pools supply no feasibility contrast and should not dominate training. Network weights update only **offline**; a verifier rejection at deployment only triggers fresh proposals.

Repaired teaching case `J02`. Use two genuine direct blockers A, B and two candidate vacated/available physical spots P, Q; no other direct blockers are present. Confirm with oracle that $A \to P, B \to Q$ and $A \to Q, B \to P$ are both supported, target-clear and pair-nonoverlapping; then and only then mark both terminal-valid. Include all-stay, one-move, and duplicate-destination negatives. This is an **isolated two-blocker control**, NOT dense `T02`. A dense five-blocker analog must explicitly assign the other three blockers and hold those placements fixed across all pool comparisons; verify every candidate independently. Neither assignment is presumed executable until full motion/order replay succeeds.

10. Search and solver algorithms

For a live query, always first verify the unchanged target with a connector from actual arm state. If already feasible return zero **additional** moves and record cumulative $|H|$; no unnecessary relocation. Otherwise try increasing total-budget values $b \geq |H|$, budget for new objects $b - |H|$. Generate candidate assignments via analytic A0, categorical Boltzmann, greedy, or uniform controls; deduplicate; check endpoints, then attempt orders, paths, connectors and full target replay. Preserve the state/hash and moves throughout. Do not restrict candidate objects only to predicted direct blockers: indirect relocation dependencies must remain selectable.

Use a memoized verifier key containing actual geometry/robot pose, moved history, remaining budget, trajectory hash, contact policy and planner settings. Do not cache a stochastic `UNRESOLVED` as definitive infeasibility. At fixed call/time budgets, compare *proposal quality* separately from *kinematic verifier bottlenecks*. A proposal that passes terminal screening but fails order/motion is **not** a false terminal label.

11. Dataset and fixture contracts

Families retain definitions from v1.3, with strengthened feasibility provenance:

Family	Required diagnostic and qualification
F0	Already feasible; zero-move result after complete arm check
F1	Exactly one direct frozen-sweep blocker; verify one corrective relocation
F2	Two or more independent direct blockers; per-object conflict contribution changes can coexist with unchanged global action failure
F3	Pairwise endpoint competition: two assignments to the same cell invalid, alternatives valid
F4	Occupied desired endpoint: a particular choice is unavailable; distinguish genuinely **necessary executable** dependency vs an avoidable assignment
F5	New obstruction on another frozen-sweep phase after candidate relocation
F6	Multiple oracle-valid alternative terminal repairs; do not label only teacher's choice positive
F7	Near but nonintersecting, physically irrelevant objects
F8	**Restricted-domain-only** infeasibility with exact finite certificate, plus mandatory unrestricted twin
F9	Terminal-valid but order/motion blocked, unverified or search-unresolved; strict status distinction
F10	Held-out mixtures of trajectory classes, access dependencies and blocker counts

For every root save stable IDs, family ancestry, reference **populated** scene and selected trajectory-bank hashes, all changed-state fields, intervention provenance, masks/images with calibration, learner/privileged namespaces, legal domain and all-supported control, local/joint labels, true direct blockers and dependencies, full motion/order outcomes, replay/attachment scope, proof lower/upper bounds and uncertainty reason. Retain incomplete-but-honest evidence. Split **before** interventions, re-renders and hard-negative mining by independent root. Never allow sibling target queries into different splits. Compare 2D (30–40) to 3D (6–10) as distinct density distributions. Use small guaranteed-scoped development roots first, not only five-direct-blocker cases.

Scale is explicit: 2D development **0 → 1 → 2** direct blockers, then 20 audited dense roots, then 300 independent roots (210/45/45 by root) for throughput, then a conditional 1,000-**root** pilot if robot/oracle throughput supports it. If an expansion target is stated in *queries*, count it separately: $Q = \sum_r Q_r$ sibling action queries, $J = \sum_q M_q$ terminal pool assignments. Do not call 1,000 sibling queries "1,000 independent roots." 3D follows 20 audited / 300-root / conditional expansion with separate counts.

The older 1,000-root methodology target and 1,000-base-query guide target are harmonized here to **a possible 1,000 independent root expansion**, with actual query count declared, not assumed.

12. Evaluation, comparisons and ablations

Use **different denominators**:

$$\text{BlockedRepairSuccess} = \frac{\#\{\text{eligible blocked queries with verified corrective replay}\}}{\#\{\text{all eligible blocked queries evaluated}\}},$$

$$\text{OverallQuerySuccess} = \frac{\#\{\text{eligible queries ending in verified target replay, including already-feasible queries}\}}{\#\{\text{all eligible queries evaluated}\}},$$

$$\text{CounterfactualValidity} = \frac{\#\{\text{claimed repairs with verified target restoration}\}}{\#\{\text{claimed repairs}\}},$$

and unnecessary-move rate on already-feasible queries. Explicitly report zero-denominator metrics as N/A, unresolved and out-of-scope counts, scoped minima (or intervals), median/tail latency, terminal pass rate, unique proposals, full order/planner calls, solver training/inference time, dynamic replay success and scripted multi-step task success. Bootstrap confidence intervals **over independent roots**, not correlated twins. Compare under identical geometry, legal domain, verifier, callback budgets and where possible wall-clock budget.

ID	Variant	Isolated scientific question
A0	Exact geometry + min-cost matching/k-best + same full motion/order verifier	Strongest permitted-geometry direct-search baseline; use general solver when matching assumptions fail
A1	Exact oracle local factors + same categorical Boltzmann sampler	Effect of learned factor error, with proposal mechanism held fixed
A2	Uniform/random legal assignments + verifier	Generic proposal structure control
A3	Learned geometry-only unary + pair, local training then joint	Do learned factors/proposals help without appearance?
A4	Full learned visual+geometry local+joint	Incremental value of visible object features
A4-L	Same full visual+geometry model, **local-only** checkpoint	Isolate benefit of joint loss for final visual model
A5-I	Full pair-trained model with pair factors ablated **only at inference**	Pair-energy inference lesion, not retraining claim
A5-R	**Retrained unary-only**, same matched training queries, parameters/compute recorded	Genuine pair-supervision/representation ablation
A6	Learned same energies + greedy instead of Gibbs	Sampling versus greedy search

ID	Variant	Isolated scientific question
A7	Learned model with envelope channel removed (and no equivalent leak)	Contribution of action geometry conditioning
A8	No coupled joint training examples **with matched total label budget**	Coupled example utility (do not conflate with loss ablation)
A9	Joint examples present, $\lambda_J = 0$ and both full/geometry-only checkpoints	Incremental contribution of joint objective
A10	Controlled one-edit twins vs **matched observational** training with equal roots, labels, family mix	Impact of matched interventions, independently of sample-count changes
A11	All-supported vs explicit restricted candidates, matched reference roots	Whether pairwise/repair difficulty is genuine or induced by domain rules

A0, A1, A2, A3 and A4 are baseline minimum. A4-L, A5-R, A8, A9, A10 are **required before claims** about pairwise learning, joint supervision or controlled interventions. A5-I is a lesion and cannot substitute for A5-R. An advantage of A8 over an observational set with more/less label coverage is confounded; report coverage and variance. Also perform geometry-only vs full on identical cylinders and reroute-first without relocation. If A0 wins with known geometry, report that honestly: correctness does not imply learned planning advantage.

13. Long-horizon integration as a required separate gate

Use an initial scripted sequence of at least **three** original target actions a_1, a_2, a_3 with 2–3 action-specific *prescribed and hashed* trajectories. **Specify when each was planned/frozen**. Two valid test protocols: (P1) *pre-frozen sequence*: plan all in a declared reference/predicted state, freeze before repairs and evaluate each later action on the achieved scene; (P2) *just-in-time frozen*: plan the next action after the preceding actual action finishes, freeze it before any repair interventions for that action. Report which protocol was used; never compare them without labeling the distinction.

Construct a scripted case where a repair of a_1 succeeds locally but makes a_3's originally pre-frozen trajectory collide. Continue from the **achieved** scene, not a reset reference. Preserve history H **within** each repair episode; reset it only after that target episode succeeds or is expressly abandoned, not between individual corrective actions. This exposes the limitation of next-action local repair; no plan-suffix energy or global task-optimality claim is introduced. Report local repair success and complete script success separately, and log the collision/causal contributions of any future blockage.

14. Acceptance and falsifiability conditions

1. Every accepted root has a populated and independently verified complete 3R or xArm7 target skill, plus exact immutable action hash; no disc-derived final labels.

2. Exact analytic unit tests reproduce contact, clearance, grid pitch, pair symmetry, boundary trivalues, uninflated/inflated equivalence and conservative intervals.
3. Restore-to-reference must **always** pass the terminal oracle in qualifying all-supported identical-cylinder intervention roots, and matching must return $|D|$; this is a *regression test* and a scientific diagnostic of benchmark design.
4. Every restricted no-solution fixture contains an actual finite unsatisfiability witness, not a no-path timeout.
5. For every `MINIMUM_CERTIFIED`, a sound lower bound equals complete independently verified upper-bound witness under the **same** scope.
6. No 3D report counts contact-disabled/gravity-defying attachment as unassisted execution; height and upright constraints enforced on all intervals.
7. All benchmark families and negative controls have stage-backed figures and machine-readable failure reason; no invented results.
8. Compare the analytic direct solver on identical known geometry and domains. If no benefit for learning is measured, restrict the research claim accordingly.

15. Exact limitations

This study is action-specific, finite-domain and move-once. It assumes independently supported cylinders, explicitly supplied geometry, GT object masks and a validated reference robot action. No results are claimed for unsensed objects, belief surprise, fluid pouring, arbitrary articulated obstacle transformations, unknown 3D shape, latent causal variable identifiability, unconstrained global TAMP completeness, or guaranteed optimal motion under incomplete randomized planning. The representation backbone, factor graph and Boltzmann sampling are established component families; proposed novelty lives in their **intervention-supervised, executable, carefully evaluated composition**, if ablations show it.

Companion: *Latent Causal TAMP — Implementation and Experimental Guide v1.6*, same date. Its stage registry `G0-G10` is the **only** milestone numbering; legacy `M6/M7` labels must not be reused.